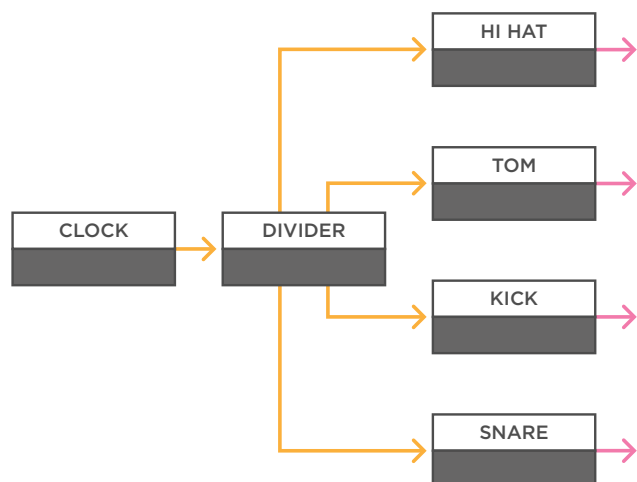


Clock Divider

Percussion

— Audio
— Control Voltage
— Trigger / Gate

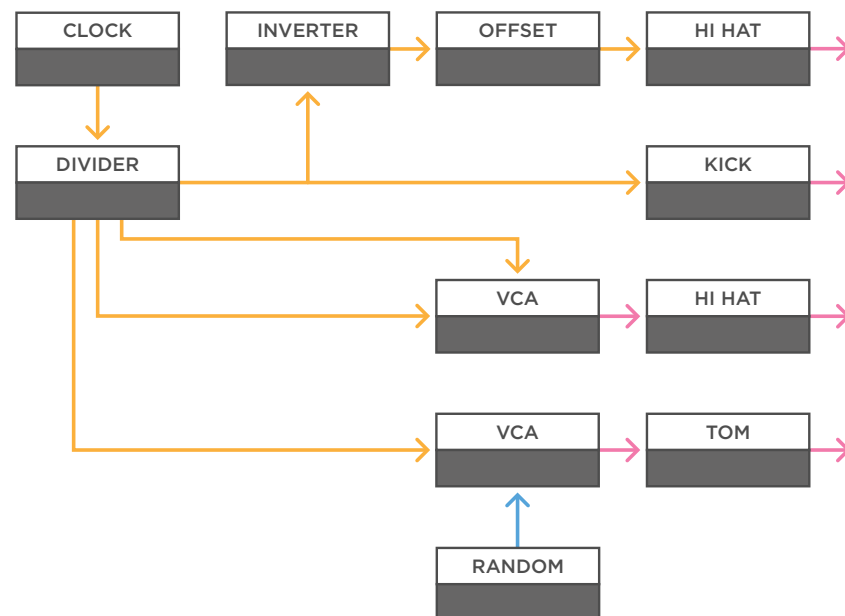


(a)

a - Let's start with a basic patch, and use a single clock and divider to drive a full set of percussive sounds. For example, if you take a single modular clock, like a square wave LFO, and send it to a clock divider, you immediately have a range of divisions to trigger percussive modules. You can send different divisions to a hi-hat, tom, kick, and snare for example.

MONOTRAIL

TECH TALK



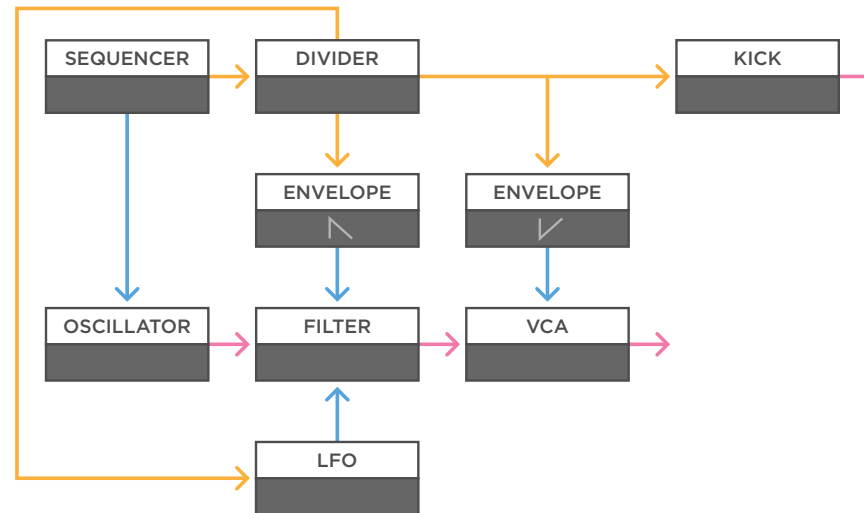
(b)

b - This patch makes clever use of a few basic utilities to create variations to the basic clock divisions. Again, there's a clock driving a clock divider. One division is triggering a kick, and a multiplication of that signal is sent through an inverter and mixed with an offset voltage, in order to create a trigger on the up-beat, for the open hi-hat. A fast division is fed into a VCA, opened by a slower division, to periodically trigger faster hats. And finally another division is fed into a VCA opened by a random voltage to trigger a high tom.

Clock Divider

Synth Voice

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - Within a modular system, triggers and gates can be used on much more than drum modules. For example to create rhythmic modulation in a synth voice with an oscillator, filter and VCA. If you use a sequencer to create a melody, and use the clock of that sequencer to drive a clock divider, you create a world of options.

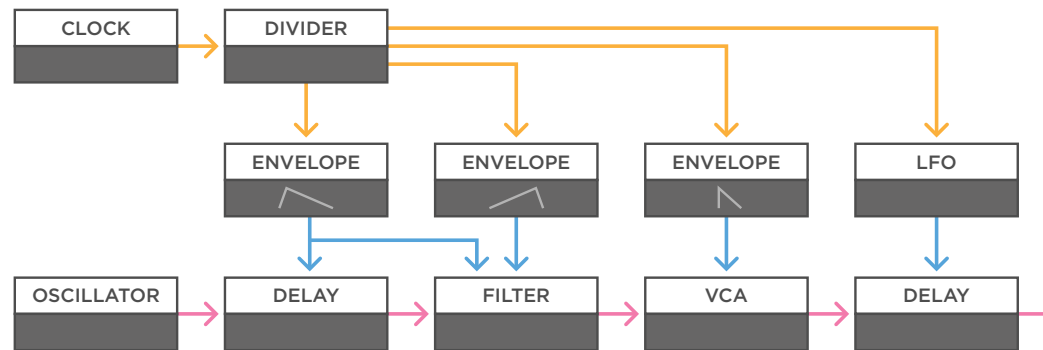
You could take a fast clock division to trigger an envelope to open the filter for some basic dynamics. But for example, you can also take a slower division to trigger a kick, multiply that signal to a short inverted envelope and send it to the VCA, to duck the synth voice every time the kick hits. In this patch an even slower division is used to sync a sine wave LFO, modulating the filter as well.

Clock Divider

Synth Voice

- Audio
- Control Voltage
- Trigger / Gate

MONOTRAIL TECH TALK



(a)

a - Here you see a slightly more expanded voice with an oscillator, delay, filter, VCA, and finally another delay, set up to create a steady drone. Again, you can use a clock and clock divider to get creative with tempo synced modulation. For example, you can use a quick division on a plucky envelope opening the VCA. A slightly longer envelope triggered by a slower division to modulate the first delay, and send a copy of that signal to the filter.

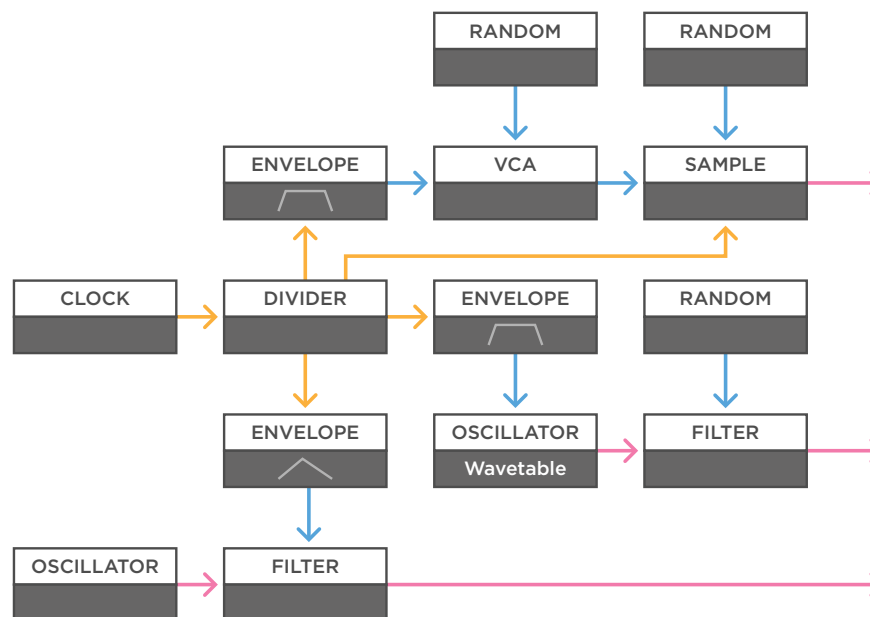
Then, add a slowly rising envelope, triggered by an even slower division, and send that to the filter as well. Finally, there's an LFO, tempo synced to another division to modulate the second delay.

Clock Divider

Generative

- Audio
- Control Voltage
- Trigger / Gate

MONOTRAIL TECH TALK



(a)

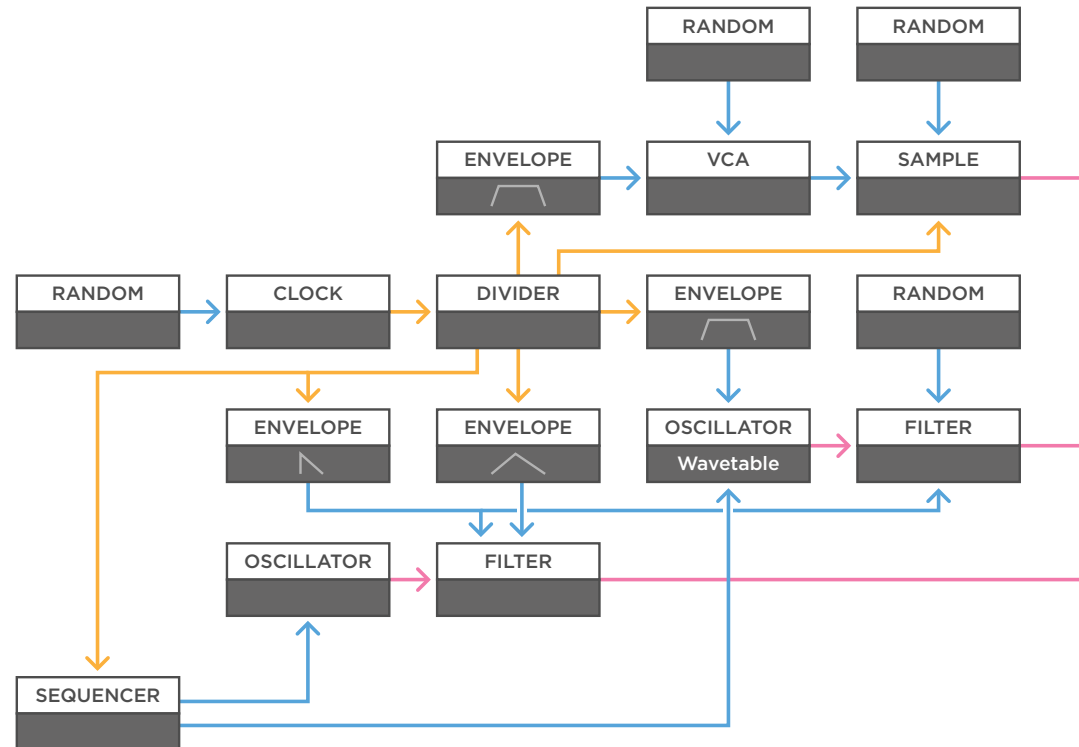
a - Clock dividers are also great as the heart of more involved generative or drone patches. Let's set up a more complex patch, based around a single clock and clock divider. To create some percussion you can use a single sample player with multiple hits. Have a faster clock division trigger the sample player, and use a random voltage to the sample select, to create random tempo synced drums. To add some more dynamics, you can send a slower division to gate an attack hold decay envelope, and have that open a VCA, so another random voltage can occasionally pass on to modulate the pitch of the sample player.

To get some melodic parts going there are two simple voices, with just an oscillator and filter. The first voice has a random voltage modulating the filter, and a slower clock division is used to gate another attack hold decay envelope, which is modulating the wave shape on the wavetable oscillator. Another slow division is used to trigger a long attack decay envelope, which is modulating the filter of the second voice, effectively fading it in and out over time.

Clock Divider

Generative

- Audio
- Control Voltage
- Trigger / Gate



(a)

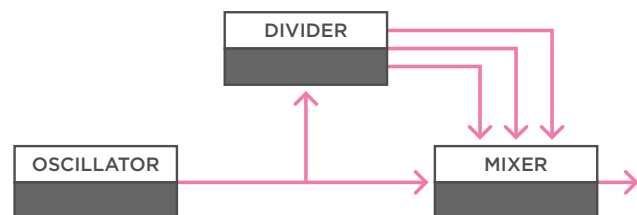
a - Finally, a fast division is used on a shorter envelope. The signal of that envelope is multiplied and sent to the filters of both voices. The same fast division is also used to clock a sequencer, in this case the Korg SQ-1, so there are two independent sequences available to create melodies for the oscillators in both voices.

The clock divider is the one module in this patch that orchestrates all the movement. And of course, you don't have to send it a steady clock. If you use this same patch, but add a slow random voltage to influence the core clock speed, you suddenly end up in lovely freestyle generative music, with a slowly drifting clock.

Clock Divider

Bonus Tricks

- Audio
- Control Voltage
- Trigger / Gate

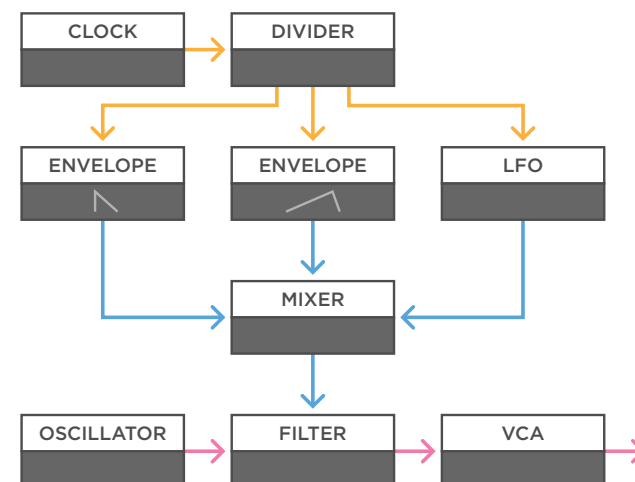


(a)

a - Because analog clock dividers respond really well to audio rate signals, you can also use them to create multiple sub oscillators. If you feed a regular square wave from an oscillator to a mixer, as well as a clock divider, you can use multiple division to the same mixer to create huge sounds. If you use regular divide by 2 or 4 divisions you get signals one or two octaves below your original frequency. But if you have a clock divider that divides by other ratios, like three or seven, you can create sub divisions with different intervals.

MONOTRAIL

TECH TALK



(b)

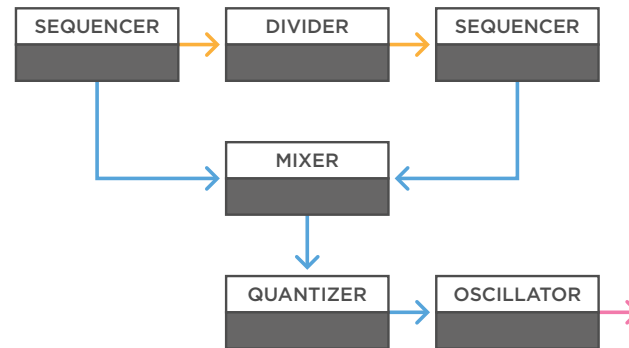
b - If you are performing with your modular it can be a great trick to use a clock and divider to trigger a variety of modulation sources. For example, one fast division to a plucky envelope, a slower division to a rising envelope, and an even slower division to sync an LFO. Then use a mixer to mix all those modulation sources together, and then send the mix off to something like a filter in a simple voice. Now you can easily create transitions and dynamics in your track or performance by mixing the different signals over time by hand. In this example the mixed signal is sent to a quantizer, and then an oscillator, to create looping melodies.

Clock Divider

Bonus Tricks

- Audio
- Control Voltage
- Trigger / Gate

MONOTRAIL TECH TALK



(a)

a - Clock divisions can also be used to drive sequencers or other pattern generators. For example if you use one sequencer as your master clock, and send it to a divider, you can use a slower division to drive another sequencer. Of course, you can use CV sequences on anything you want, but just to give an example: You can use the first sequencer to create a 7 step melody, send that to a mixer, a quantizer and then the oscillator in a synth voice. Then you can use the second slower sequence to add longer sustained voltages to the first sequence, to create variations.